

Numerical and Computational Methods

Course Name: Numerical and Computational Methods

Course Code: MC 225

Course Credit: 4 (3 Lecture hours (3-0-2-4))

Prerequisite: Calculus, Linear Algebra

Course Objective: The objective of this course is to make the students learned some numerical techniques and implements them to find more appropriate solution to their problems in Engineering and Data Science.

Course Contents:

1. Numerical Solution of Algebraic and Transcendental Equations: Polynomials, Bi-section Method, Secant method, Newton Raphson method, Method of Successive Approximations.
2. Numerical Solution of Systems of Linear and Non-linear Algebraic Equations: Gauss Elimination method, Gauss Jordan method, Gauss-Seidel iteration method, Jacobi iteration method, Newton's method (for non-linear equations).
3. Interpolation: Lagrange interpolation, Difference table, divided difference interpolation, Newton's Forward and backward difference interpolation, piecewise interpolation, Spline interpolation.
4. Differentiation and Integration: Numerical differentiation using Lagrange interpolation, using Difference operators and using the method of undetermined coefficients.
Numerical Integration: Trapezoidal rule, Simpson's rule and Gaussian quadrature formulae.
5. Numerical Solution of Differential Equations: Introduction, Euler's method, Midpoint rule, Second-order Runge—Kutta method, Fourth-order Runge—Kutta method, Applications to Boundary value problems. MATLAB for the solution of ordinary differential equations.

Reference Books:

- Introductory Methods of Numerical Analysis, S. S Sastry, PHI, New Delhi.
- Numerical Analysis, Francis Scheid, McGraw Hill — Schaum's Outlines, New York.
- Numerical Methods for Scientists and Engineers, K. Sankara Rao, PHI, New Delhi.
- Numerical Methods using MATLAB, J. H. Mathews, K.D. Fink, Pearson.
- Elementary Numerical Analysis, K. Atkinson & W. Han, Wiley-India, New Delhi.
- Numerical Methods for Engineers and Scientists, J. D. Hoffman, Marcel Dekker Inc., New York
- An Introduction to Numerical Analysis, K. E. Atkinson, Wiley-India, New Delhi
- An Introduction to Numerical Methods and Analysis, J. F. Epperson, Wiley, New Jersey

Course Outcomes:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X		X	X							X

Course Schedule:

Sl. No.	Topics	No. of lectures
1	Numerical Solution of Algebraic and Transcendental Equations: Polynomials, Bi-section Method, Secant method, Newton Raphson method, Method of Successive Approximations.	6

2	Numerical Solution of Systems of Linear and Non-linear Algebraic Equations: Gauss Elimination method, Gauss Jordan method, Gauss-Seidel iteration method, Jacobi iteration method, Newton's method (for non-linear equations).	8
3	Interpolation: Lagrange interpolation, Difference table, divided difference interpolation, Newton's Forward and backward difference interpolation, piecewise interpolation, Spline interpolation.	8
4	Differentiation and Integration: Numerical differentiation using Lagrange interpolation, using Difference operators and using the method of undetermined coefficients. Numerical Integration: Trapezoidal rule, Simpson's rule and Gaussian quadrature formulae.	8
5	Numerical Solution of Differential Equations: Introduction, Euler's method, Midpoint rule, Second-order Runge—Kutta method, Fourth-order Runge—Kutta method, Applications to Boundary value problems. MATLAB for the solution of ordinary differential equations.	10

Tentative Evaluation Policy:

1. First In-Semester Exam (25%)
2. Second In-semester Exam (25%)
3. End Semester Exam (40%)
4. Lab Work (10%)